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Robust Beta Estimation: A Comparative Analysis of OLS and RANSAC

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Abstract

This paper investigates the sensitivity of beta estimation to outliers and market regime shifts, combining controlled simulations with real financial data from GOOG, TSLA, and the S&P 500 index. Traditional Ordinary Least Squares (OLS) regression is compared to a robust alternative based on the RANSAC algorithm, which identifies and removes anomalous observations before estimating the slope. Simulation results show that even moderate contamination (10% outliers) leads OLS to substantially underestimate or overestimate the true beta, while RANSAC consistently recovers the correct value. When contamination is minimal (0.25%), the two estimators converge, indicating that data irregularities—not model structure—drive the divergence. Empirical evidence confirms these findings. GOOG displays limited differences between OLS and RANSAC, reflecting its stable return profile, whereas TSLA exhibits substantial discrepancies caused by its high volatility and frequent extreme observations. A regime analysis further highlights that OLS becomes unreliable during crisis periods, while RANSAC maintains stability by filtering out shock-driven distortions. Overall, the results demonstrate that robust methods provide more reliable measures of systematic risk, especially in volatile markets or for assets with heavy-tailed return distributions. The study emphasizes that beta is not only asset-specific but also regime-dependent, and that combining classical and robust estimators yields a more accurate understanding of market exposure.

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1 Introduction

In financial economics, the beta (β) coefficient measures how sensitive a security's return is to movements in the overall market. It is a cornerstone of the Capital Asset Pricing Model (CAPM), which links the expected return of an asset to its exposure to systematic risk.

According to the CAPM:

$$E(R_i) = R_f + \beta_i (E(R_m) - R_f)$$

Where:

- $E(R_i)$ = expected return of asset i
- R_f = risk-free rate
- $E(R_m)$ = expected return of the market portfolio
- β_i = sensitivity of asset i to market fluctuations

A beta greater than 1 indicates that the asset tends to amplify market movements (higher volatility), while a beta lower than 1 suggests a more defensive behavior. For example, if $\beta = 1.2$, the stock is expected to rise or fall by 1.2% for every 1% movement in the market.

1.1 OLS Estimation of Beta

In empirical analysis, β is typically estimated using Ordinary Least Squares (OLS) by fitting the linear regression:

$$R_i - R_f = \alpha + \beta(R_m - R_f) + \varepsilon$$

where both the asset and the market are expressed in excess returns over the risk-free rate.

The OLS estimator of beta is:

$$\hat{\beta} = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

This method implicitly assumes:

- a stable linear relationship between asset and market,
- normality of returns,
- absence of extreme observations that distort the estimation.

1.2 The Problem: Financial Returns Are Not Normal

Real-world financial markets often violate these assumptions. Returns typically exhibit:

- fat tails,
- outliers,
- jumps caused by news, crises, or rare events.

Such extreme values can heavily influence the OLS estimate, making β :

- unstable,
- biased,
- or even completely misleading.

1.3 Motivation for Robust Beta

To address these limitations, this project explores the concept of a *robust beta* — a measure of systematic risk designed to remain stable and reliable even in the presence of extreme market events.

By applying robust estimation techniques, we aim to capture the true sensitivity of a stock to the market without being distorted by short-term anomalies, crisis periods, or occasional outliers.

2 Explanation of Robust Measures

In financial data analysis, returns frequently contain outliers — extreme observations produced by market shocks, news events, liquidity crises, or data errors. These values can significantly distort the results of a traditional Ordinary Least Squares (OLS) regression, because OLS minimizes the sum of squared residuals, giving disproportionate weight to large errors.

Formally, OLS solves:

$$\min_{\alpha, \beta} \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$$

This quadratic loss makes OLS highly sensitive even to a single anomalous point. To overcome this weakness, we turn to robust regression methods — estimators designed to remain stable when:

- the data are not normally distributed,
- returns contain shocks or spikes,
- extreme events influence only part of the sample.

Below we present the three main robust estimators considered in this project.

2.1 Huber Regression

The Huber estimator is a compromise between OLS and Least Absolute Deviations (LAD). It applies:

- a quadratic penalty to small residuals (like OLS),
- a linear penalty to large residuals (like LAD).

This limits the influence of extreme values while keeping smooth estimation for regular points.

The loss function is:

$$\rho(u) = \begin{cases} \frac{1}{2}u^2, & |u| \leq \delta \\ \delta(|u| - \frac{1}{2}\delta), & |u| > \delta \end{cases}$$

where δ is a threshold determining when the transition from quadratic to linear penalty occurs.

Huber regression is:

- smoother than LAD,
- less sensitive than OLS,
- but still partially affected by large outliers.

It is ideal when the dataset contains moderate contamination, not extremely heavy tails.

2.2 Theil–Sen Estimator

The Theil–Sen estimator is a non-parametric, rank-based method. It estimates the slope as the median of all pairwise slopes:

$$\hat{\beta}_{TS} = \text{median} \left(\frac{y_i - y_j}{x_i - x_j} \right), \quad i < j$$

The intercept is then computed as:

$$\hat{\alpha}_{TS} = \text{median}(y_i - \hat{\beta}_{TS}x_i)$$

Key features:

- extremely robust to outliers,
- invariant to up to 29% arbitrary corruptions,
- captures a central tendency of the relationship,
- slower on large datasets due to the combinatorial number of slopes.

Theil–Sen works very well when outliers are isolated or sparse, but may struggle when extreme events appear in clusters, as in financial crises.

2.3 RANSAC (Random Sample Consensus)

We adopt RANSAC as the primary robust measure of beta in this project. RANSAC differs from the previous methods because it does not smooth outliers or reduce their weight — it actively identifies and excludes them.

How RANSAC works

RANSAC iteratively:

1. randomly selects a small subset of the data,
2. fits a regression to that minimal subset,
3. tests all remaining points, classifying them as:
 - **inliers:** points consistent with the model,
 - **outliers:** points too far from the model,
4. repeats this process many times and keeps the model with the largest number of inliers.

The final regression is then estimated only on the inlier set, ignoring all other observations.

Why RANSAC is ideal for financial data

Financial returns tend to show:

- fat tails,
- clusters of extreme values,
- shocks persisting for multiple days,
- non-Gaussian behavior during crisis periods.

RANSAC is particularly suited for such data because:

- it tolerates high percentages of outliers (10–20% or more),
- it can identify and discard clusters of abnormal points,
- it produces a crisis-resistant beta driven by the stable majority of observations,
- it does not rely on strong distributional assumptions.

Compared to Huber and Theil–Sen, RANSAC offers the highest level of robustness because it removes outliers entirely rather than down-weighting them.

2.4 Conclusion: Why We Choose RANSAC Beta

Robust regression techniques allow us to estimate a beta that is:

- more stable,
- less sensitive to extreme events,
- closer to the true systematic risk of an asset.

Among the alternatives, RANSAC provides the strongest protection against outliers, especially when contamination is large, non-Gaussian, and clustered — a typical characteristic of real financial time series.

For these reasons, RANSAC will be used as the primary robust beta estimator throughout the empirical analysis.

3 Bias–Variance Tradeoff

The bias–variance tradeoff is a fundamental concept in statistical learning and regression analysis. It describes how the expected prediction error of an estimator can be decomposed into two key components: bias and variance. Understanding this tradeoff is essential when comparing OLS with robust regression techniques, because these methods exhibit systematically different behaviors under noisy or contaminated data.

Bias

Bias measures how far, on average, an estimator’s predictions are from the true underlying relationship. A low-bias estimator captures the structure of the data well, while a high-bias estimator systematically deviates from the true model. Robust regression methods typically introduce a small amount of bias, because they do not fully trust all observations—especially those far from the central trend.

Variance

Variance captures the sensitivity of an estimator to fluctuations in the sample. A high-variance estimator reacts strongly to small changes in the data, producing unstable estimates across samples. OLS, because it gives quadratic weight to residuals, tends to have higher variance in the presence of outliers or heavy-tailed returns.

Tradeoff

The bias–variance tradeoff states that:

- Methods like OLS have **low bias** but can suffer from **high variance** when the data contain shocks, jumps, or anomalous observations.

- Robust methods introduce a small amount of **bias**, but significantly reduce the **variance**, leading to more stable estimates.

In financial contexts, minimizing variance is often more important than minimizing bias. Stability is crucial, especially when regression estimates feed into portfolio construction, risk budgeting, or hedging strategies.

The following figure illustrates these concepts: OLS produces estimates that are unbiased but disperse widely in noisy data, whereas robust regression yields more concentrated predictions at the cost of a slight displacement from the target.

Bias-Variance Tradeoff

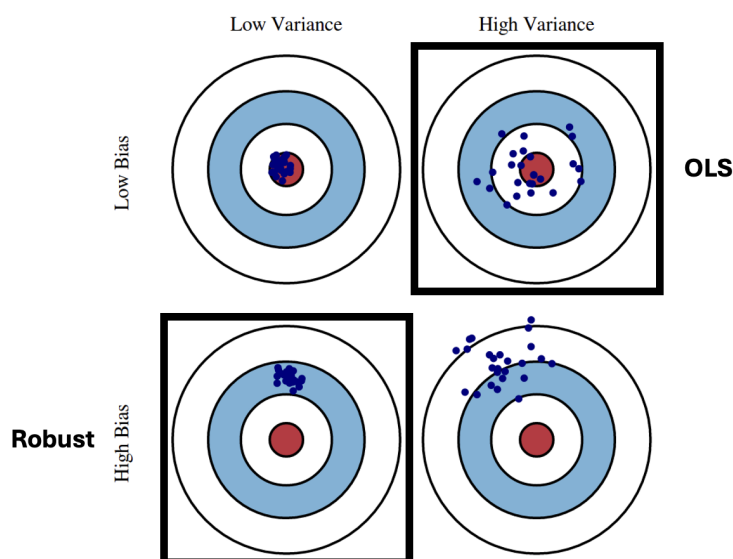


Figure 1: Graphical intuition of the bias–variance tradeoff for OLS and robust estimators. OLS maintains low bias but suffers from high variance in the presence of anomalies. Robust methods exhibit slightly higher bias but significantly lower variance.

4 Why Accurate Beta Measurement Matters

Estimating beta incorrectly has concrete financial implications. Beta is frequently used in portfolio hedging, factor-neutral construction, capital allocation, and regulatory risk assessment. A biased or unstable beta leads to strategies that appear neutral in-sample but behave unexpectedly during market stress.

The following illustration provides a clear example. A fund manager wants to build a beta-neutral position on TSLA using long and short exposures. The objective is:

$$\beta_{\text{portfolio}} = 0$$

If the *real* beta of TSLA differs from the *model-implied* beta—such as an OLS estimate distorted by outliers—the resulting hedged position will no longer be neutral. The portfolio may unintentionally take on systematic risk, generating losses during market movements.

In the example:

- The real beta of the long position is higher than the model suggests.
- The beta of the short leg is also misestimated.
- When the market declines, the supposedly hedged portfolio produces a non-zero loss.

This demonstrates why a robust and correctly estimated beta is essential for risk management. Robust procedures—such as RANSAC—avoid underestimating or overestimating exposures by removing crisis-induced outliers that distort OLS results.

The figure below visualizes this intuition.

<u>Motivation</u>		Comparison: β_{real} vs. β_{model}	
\$100M long on TSLA		Constraint: $\beta_{\text{Portfolio}} = 0$	
Real	$\beta_{\text{long}} = 2.5$	$\Rightarrow$	$\beta_{\text{short}} = -1$
	$\beta_{\text{short}} = -1$		
		$\Rightarrow$	
		Market -5%	
		$\Rightarrow$	
		Result = \$0	
Model	$\beta_{\text{long}} = 2$	$\Rightarrow$	$\beta_{\text{short}} = -1.25$
	$\beta_{\text{short}} = -1.25$		
		$\Rightarrow$	
		Market -5%	
		$\Rightarrow$	
		Result = -\$4.5M	

Figure 2: Motivational example: comparison between real and model-implied beta in a TSLA hedging strategy. Misestimated beta causes the portfolio to deviate from neutrality, generating losses when the market moves.

5 What Should We Expect?

Before analysing real financial time series, it is useful to understand how the methodology behaves in a controlled environment. To do this, we generate synthetic data where the true beta is known, allowing us to compare:

- the traditional OLS beta, which is highly sensitive to outliers,
- the RANSAC beta, designed to remain stable even under contamination.

The simulation framework is based on two dimensions:

1. Comparison between two synthetic stocks:
 - Stock A with 0.25% outliers,
 - Stock B with 10% outliers.
2. Comparison between two market regimes:
 - a calm, low-volatility period,
 - a crisis, high-volatility period with heavy contamination.

5.1 Simulating Synthetic Financial Data

Each simulated dataset contains:

- 2500 daily observations,
- a true beta equal to 1.50,
- market return volatility of 1.2%,
- idiosyncratic stock volatility of 1.0%.

The uncontaminated returns follow the simple linear model:

$$y = \beta \cdot x + \varepsilon,$$

where x is drawn from a normal distribution with volatility σ_x and ε is random noise with volatility σ_ε . This represents an ideal situation in which the stock moves proportionally to the market, plus idiosyncratic noise.

5.2 Introducing Outliers

Financial returns often contain irregular and extreme observations. To replicate these conditions, the simulation introduces two types of anomalies.

1. Vertical Outliers in Stock Returns (y)

These mimic firm-specific shocks:

- earnings surprises,
- sudden jumps,
- idiosyncratic events.

They are generated by adding unusually large noise to random observations of y .

2. Leverage Outliers in Market Returns (x)

These represent:

- macroeconomic surprises,
- liquidity events,
- abnormal market behaviour.

They are produced by injecting large values into random points of x .

The total proportion of outliers is divided as follows:

$$75\% \rightarrow \text{vertical outliers in } y, \quad 25\% \rightarrow \text{leverage outliers in } x.$$

Two stock scenarios are generated:

- Stock A: very small outlier contamination (0.25%),
- Stock B: heavy contamination (10%).

This allows us to study how increasing levels of contamination distort beta estimation.

5.3 Why Outliers Matter

Outliers strongly distort the OLS regression line. OLS minimizes:

$$\sum_{i=1}^n (y_i - \beta x_i)^2,$$

which gives disproportionately large weight to extreme residuals. As a result, outliers can produce:

- biased beta estimates,
- a rotated regression line,
- misleading conclusions about systematic risk.

The synthetic experiment allows us to observe exactly how OLS reacts to contamination and how RANSAC prevents such distortions.

5.4 Calm vs. Crisis Scenarios

In addition to stock-specific contamination, we simulate two market regimes:

Calm Period

- low volatility,
- very few outliers (0.5%),
- stable relationship between stock and market.

Crisis Period

- volatility increases more than six-fold,
- heavy contamination (15% outliers),
- two types of extreme events:
 - **extreme negative stock shocks** between -15% and -5% ,
 - **leverage outliers** with abnormally large values in both x and y .

The underlying model remains:

$$y = \beta \cdot x + \varepsilon,$$

but the crisis environment becomes much more unstable, enabling a clear comparison of OLS and RANSAC under stress.

5.5 Why the Simulation Focuses on RANSAC

The main objective of the simulation is to evaluate whether RANSAC provides a more reliable and crisis-resistant estimate of beta.

RANSAC is particularly well-suited for financial data because it:

- identifies and removes outliers instead of down-weighting them,
- remains stable even when a large share of observations is contaminated,
- isolates the core consensus relationship between stock and market,
- captures the true systematic risk without being distorted by rare anomalies.

This property is essential in finance, where extreme values and heavy tails are common. The simulation therefore acts as a controlled laboratory that demonstrates why a robust estimator such as RANSAC is indispensable both in normal conditions and during crisis periods.

6 Statistical Test: Comparing OLS Beta and RANSAC Beta

An important component of the analysis is a formal statistical test that evaluates whether the beta estimated through Ordinary Least Squares (OLS) differs significantly from the robust beta obtained via RANSAC. This allows us to quantify the extent to which outliers distort classical beta estimation.

The hypotheses of interest are:

$$H_0 : \beta_{\text{OLS}} = \beta_{\text{RANSAC}}, \quad H_1 : \beta_{\text{OLS}} \neq \beta_{\text{RANSAC}}.$$

6.1 How the Test Works

OLS and RANSAC both estimate the slope coefficient (beta), but they rely on different subsets of the data:

- OLS uses *all* observations, including outliers,
- RANSAC uses only inliers, explicitly discarding extreme and inconsistent points.

Because these two estimators are computed on different sample sizes and different data subsets, their exact sampling distributions are not analytically known. Nevertheless, we can still rely on the Central Limit Theorem (CLT) to perform an approximate hypothesis test.

6.2 Why a t-Test Is Appropriate

Under broad conditions, the CLT implies that:

- the sampling distribution of each estimator (OLS beta and RANSAC beta) becomes approximately normal when the number of observations is sufficiently large;
- robust estimators such as RANSAC, when based on many inliers, also converge to normality;
- the difference between two approximately normal estimators is itself approximately normal.

Therefore, the quantity:

$$\beta_{\text{OLS}} - \beta_{\text{RANSAC}}$$

is approximately normally distributed, allowing us to construct a valid t-statistic:

$$t = \frac{\beta_{\text{OLS}} - \beta_{\text{RANSAC}}}{\sqrt{\text{SE}(\beta_{\text{OLS}})^2 + \text{SE}(\beta_{\text{RANSAC}})^2}}.$$

This expression exploits the fact that the variance of the difference of two independent (or weakly dependent) estimators equals the sum of their variances.

6.3 Approximating Degrees of Freedom

Since OLS and RANSAC rely on different sample sizes, their degrees of freedom must be handled carefully. We adopt conservative approximations:

- For OLS: $\text{df}_{\text{OLS}} = n_{\text{OLS}} - 2$ (intercept + slope),
- For RANSAC: $\text{df}_{\text{RANSAC}} = n_{\text{RANSAC}} - 1$ (slope only).

To ensure the test does not overstate significance, we select:

$$\text{df} = \min(\text{df}_{\text{OLS}}, \text{df}_{\text{RANSAC}}).$$

6.4 Interpreting the Test Result

The p-value determines whether the difference between the two beta estimates is statistically significant:

- If $p < \alpha$ (typically 0.05), we reject H_0 :
 - the two betas differ significantly,
 - OLS is likely biased due to outliers.
- If $p \geq \alpha$, we do not reject H_0 :
 - the difference is not statistically significant,
 - RANSAC and OLS provide similar estimates.

The statistical test outputs a clear summary table and the full hypothesis test result. This provides a rigorous and quantitative framework for assessing whether robust regression—specifically RANSAC—offers a more reliable beta estimate in noisy or contaminated financial datasets.

7 Core Functions Used Throughout the Analysis

Before running the simulations and the empirical analysis, we import the Python libraries required for data generation, estimation, and visualisation. Each library plays a distinct and essential role:

- **NumPy**: numerical operations and random number generation, used to simulate synthetic returns.
- **Pandas**: creation and manipulation of datasets, including time-series structures.
- **yfinance**: downloading real financial data directly from Yahoo Finance.
- **Matplotlib**: graphical visualisation of scatter plots, regression lines, and comparisons.
- **SciPy** (t-distribution): performing hypothesis tests on beta differences.
- **scikit-learn RANSACRegressor**: carrying out robust regression by detecting and removing outliers.

Together, these tools provide the computational infrastructure for both the synthetic simulations and the analysis of real market data.

7.1 Function to Compute Daily Returns

To work with real financial time series, we define a utility function that downloads historical price data using **yfinance** and computes daily log-returns. This produces a clean and stationary return series suitable for regression analysis.

7.2 Residual Standard Error (RSE)

To assess the quality of each beta estimate, we compute the Residual Standard Error (RSE), a measure of how much actual stock returns deviate from the returns predicted by the model. A lower RSE indicates a tighter, more reliable fit.

7.3 OLS and RANSAC Beta Estimators

Two core estimators are implemented throughout the analysis:

1. **Classical OLS beta**, computed using the standard linear regression formulation.
2. **Robust RANSAC beta**, which identifies inliers, excludes anomalous observations, and then estimates the regression slope on the filtered dataset.

This pair of estimators allows us to compare traditional and robust approaches directly under both normal and stressed market conditions.

7.4 Plotting OLS vs. RANSAC Beta

We implement a visualisation function that plots the scatter of stock vs. market returns and overlays the regression lines for both OLS and RANSAC. The function also reports the number of outliers detected by RANSAC, providing an immediate indication of the degree of contamination in the data.

7.5 Crisis vs. Calm Beta Analysis

Another core function compares the beta of a stock across two regimes:

- a **calm period** with low volatility and minimal anomalies,
- a **crisis period** characterised by elevated volatility and heavy contamination.

For each regime, both OLS and RANSAC betas are computed, visualised, and statistically tested under the hypothesis:

$$H_0 : \beta_{\text{OLS}} = \beta_{\text{RANSAC}}.$$

This allows us to verify whether RANSAC offers a significantly more reliable beta estimate during stress episodes.

7.6 Simulating Synthetic Stock and Market Returns

Finally, we implement a function that generates artificial financial data with a known true beta. The procedure:

- creates clean market and stock return series following the linear model,
- injects controlled anomalies in the form of:

- vertical outliers (on the stock returns),
- leverage outliers (on the market returns),
- outputs a set of time series ready for testing the behaviour of OLS and RANSAC beta estimators.

7.7 Simulating Calm vs. Crisis Market Regimes

We implement a function that generates two distinct synthetic market environments: a calm period and a crisis period. The calm regime is characterised by low volatility and a minimal presence of outliers, representing a stable market environment. In contrast, the crisis regime exhibits sharply elevated volatility and a dense concentration of extreme shocks, mimicking real-world stress episodes.

This two-regime framework enables us to examine how beta estimation changes when the market transitions from a stable to a turbulent state. In particular, it allows us to evaluate whether RANSAC maintains stability and accuracy when OLS becomes unreliable due to widespread contamination in the data.

7.8 Visual Comparison of OLS and RANSAC Betas

To summarise the behaviour of OLS and RANSAC in an intuitive way, we include two plotting functions.

- **Single-Stock Comparison:** This function compares the two beta estimates for a single simulated asset. It reports the OLS and RANSAC betas, their standard errors, their deviation from the true beta, and the outcome of the formal hypothesis test

$$H_0 : \beta_{\text{OLS}} = \beta_{\text{RANSAC}}.$$

The accompanying scatter plot highlights the inliers and outliers identified by RANSAC.

- **Calm vs. Crisis Comparison:** The second function extends the analysis to the two-regime environment. It produces side-by-side scatter plots with regression lines for calm and crisis periods, illustrating how increased volatility and extreme observations affect the stability of OLS relative to RANSAC. This visualisation clearly shows that RANSAC maintains a consistent slope even under heavy contamination, whereas the OLS estimate becomes distorted.

These visual tools provide an essential complement to the statistical analysis, offering clear and interpretable evidence of how outliers and volatility regimes influence beta estimation.

8 Executing the Simulation and Comparing Robust vs. Non-Robust Betas

Having defined the computational framework and the functions used throughout the analysis, we now move to the execution of the synthetic simulations. This part of the study examines how classical OLS and robust RANSAC beta estimators behave under different market conditions, focusing in particular on the effect of outliers and extreme observations.

The simulation block performs three independent experiments, each designed to highlight specific weaknesses of OLS and the advantages of robust estimation techniques. These experiments form the empirical backbone of our analysis.

8.1 Scenario 1 — Many Outliers (10%)

In the first scenario, we generate synthetic stock and market returns where 10% of all observations are deliberately contaminated with large shocks. This setting replicates a heavily distorted market environment, allowing us to examine:

- how severely OLS beta reacts to extreme values,
- whether RANSAC is able to isolate the “clean” relationship between the stock and the market,
- the extent to which outliers rotate or bias the classical regression line.

Once the contaminated dataset is generated, we call the function `plot_scatter_comparison()`, which performs the following tasks:

- estimates the OLS and RANSAC beta coefficients,
- computes their corresponding standard errors,
- performs the hypothesis test

$$H_0 : \beta_{\text{OLS}} = \beta_{\text{RANSAC}},$$

- visualises both regression lines on a scatter plot,
- highlights the inliers and outliers detected by RANSAC.

Purpose of This Experiment

This initial simulation provides the first empirical comparison between the two estimation methods. It shows:

- how strongly OLS reacts to extreme data points,
- whether RANSAC delivers a more stable and consistent estimate of systematic risk,
- how the presence of widespread contamination affects the estimation of beta.

By observing this scenario, we establish a reference point for understanding the broader implications of robust regression in financial time series.

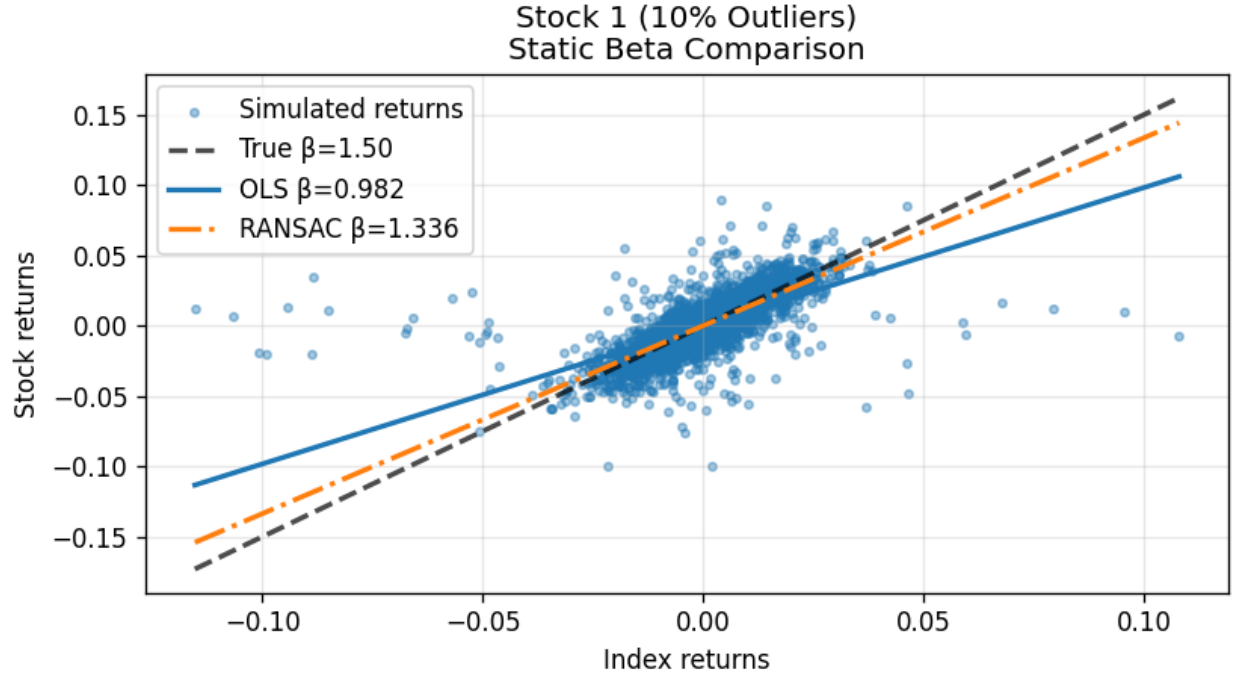


Figure 3: Static beta comparison for a synthetic stock with 10% outliers. The OLS estimate (blue line) is strongly distorted by extreme observations, whereas the RANSAC estimate (orange dashed line) remains close to the true beta (grey dashed line).

Table 1: Scenario 1 (10% Outliers): Static Beta Estimates and Hypothesis Test

Estimator	β	$\beta - \beta_{\text{true}}$
OLS	0.9820	-0.5180
RANSAC	1.3357	-0.1643

Hypothesis Test

Null hypothesis	$H_0 : \beta_{\text{OLS}} = \beta_{\text{RANSAC}}$
p-value	0.0000
Decision	Reject H_0
Conclusion	Betas are significantly different

Interpretation of Scenario 1

In this first scenario, we analyse a simulated stock whose return series has been intentionally contaminated with a large proportion of outliers (10% of all observations). The true beta used in the data-generating process is $\beta_{\text{true}} = 1.50$. This controlled setting allows us to observe how ordinary least squares (OLS) behaves when returns contain frequent and sizeable deviations from the underlying linear relationship.

Interpretation of the Figure The scatter plot reports three benchmark regression lines:

- **True relationship** ($\beta = 1.50$) The dashed black line represents the theoretical data-generating process. This is the benchmark slope that would be recovered in the absence of distortions.
- **OLS estimate** ($\beta = 0.982$) The blue line is visibly flatter than the true relationship, indicating substantial downward bias. With 10% extreme observations, OLS assigns excessive weight to anomalous points, pulling the fitted line toward them. As a consequence, OLS fails to capture the systematic connection between market and stock returns.
- **RANSAC estimate** ($\beta = 1.336$) The orange dashed line lies much closer to the true slope. RANSAC identifies and excludes points that do not conform to the dominant pattern, estimating beta using only the coherent core of the data. Although not exactly equal to 1.50, the RANSAC estimate is substantially more accurate than the OLS estimate.

This visual comparison illustrates clearly that OLS is heavily distorted by the presence of frequent outliers, while RANSAC preserves the underlying structure and yields a beta estimate closer to the truth.

Interpretation of the Numerical Results The numerical results confirm the graphical evidence:

- **OLS:** $\beta_{\text{OLS}} = 0.982$ with a deviation of -0.518 from the true beta. This constitutes a severe underestimation of systematic risk.
- **RANSAC:** $\beta_{\text{RANSAC}} = 1.3357$ with an error of -0.1643 . The standard error is also smaller, reflecting a more coherent fit to the inlier subset of the data.

Hypothesis Test To formally assess whether the two estimators differ, we test:

$$H_0 : \beta_{\text{OLS}} = \beta_{\text{RANSAC}}.$$

The resulting test statistic is $t = -12.946$, with a p-value effectively equal to zero. Thus:

- the null hypothesis is rejected,
- the two estimators are statistically different,
- OLS exhibits a significant and systematic bias induced by outliers.

This confirms that, under heavy contamination, OLS provides a misleading estimate of systematic risk, while RANSAC recovers the underlying relationship with substantially higher accuracy.

8.2 Scenario 2 — Few Outliers (0.25%)

In this second scenario, we repeat the simulation exercise under almost ideal conditions: only 0.25% of the observations are contaminated by extreme values. This setting approximates a clean financial environment, enabling us to compare the behaviour of OLS and RANSAC when the data follow the underlying linear relationship closely. Under these conditions, we expect both estimators to perform well, with minimal distortion.

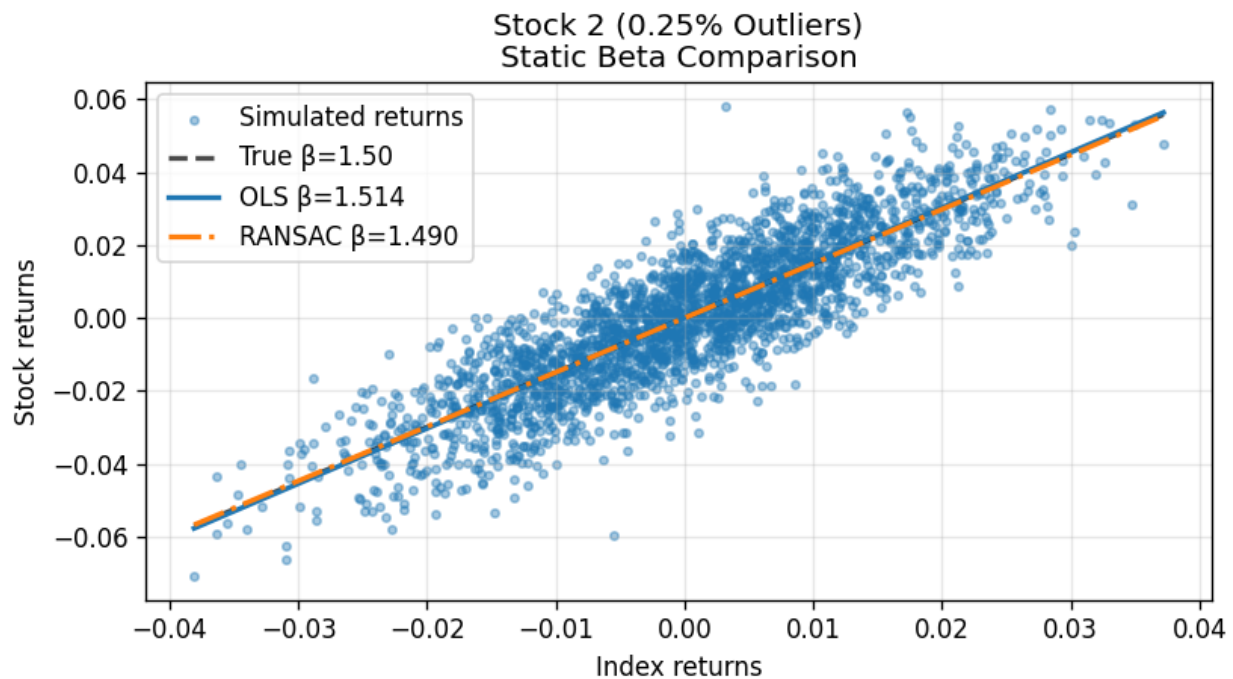


Figure 4: Static beta comparison for a synthetic stock with 0.25% outliers. In a nearly uncontaminated environment, both OLS and RANSAC remain close to the true beta (1.50), with minimal differences between the two estimators.

Table 2: Scenario 2 (0.25% Outliers): Static Beta Estimates and Hypothesis Test

Estimator	β	$\beta - \beta_{\text{true}}$
OLS	1.5145	0.0145
RANSAC	1.4895	-0.0105

Hypothesis Test

Null hypothesis	$H_0 : \beta_{\text{OLS}} = \beta_{\text{RANSAC}}$
p-value	0.2571
Decision	Do not reject H_0
Conclusion	No significant difference between the estimators

Interpretation of Scenario 2

The second scenario provides an important benchmark because the data are almost free of anomalies: only 0.25% of the observations are contaminated. Under these conditions, the simulated returns align very closely with the theoretical linear model, and the scatter plot reveals a compact and well-behaved cloud of points.

Both estimators perform as expected. The OLS beta estimate ($\beta_{OLS} = 1.5145$) and the RANSAC estimate ($\beta_{RANSAC} = 1.4895$) lie extremely close to one another and to the true value of 1.50. The small difference between the two estimators falls comfortably within normal sampling variability. This is confirmed by the hypothesis test: the p-value is well above conventional significance thresholds, and we therefore do not reject the null hypothesis that the two betas are equal.

This scenario illustrates that when the return distribution is clean and only rarely disrupted by large shocks, robust methods such as RANSAC do not introduce unnecessary corrections. RANSAC naturally converges to a result nearly identical to OLS, demonstrating that robustness does not come at the expense of efficiency when the data are well behaved.

8.3 Scenario 3 — Crisis Period vs. Calm Period

This scenario introduces a simulation with two distinct market regimes: a stable **calm period** and a highly volatile **crisis period**. These contrasting environments allow us to study how the sensitivity of stock returns to market movements (beta) changes when volatility increases and outliers become more frequent. The calm regime contains low noise and only a few extreme observations, while the crisis regime is characterised by large shocks, fat tails, and a heavy concentration of outliers.

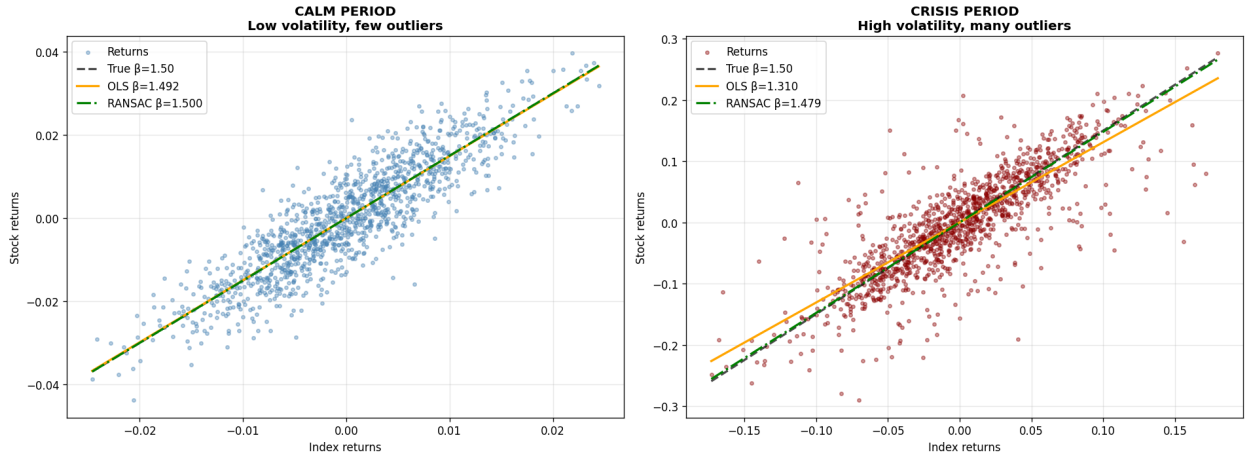


Figure 5: Static beta comparison across calm and crisis market regimes. During the calm period (left), both OLS and RANSAC estimates remain close to the true beta of 1.50. During the crisis period (right), OLS becomes significantly biased due to extreme observations, while RANSAC remains stable and close to the true slope.

Table 3: Scenario 3: Calm vs. Crisis Period — Beta Estimates and Hypothesis Tests

Calm Period		
	β	$\beta - \beta_{\text{true}}$
OLS	1.4921	-0.0079
RANSAC	1.5000	0.0000

Hypothesis Test (Calm)		
Null hypothesis	$H_0 : \beta_{\text{OLS}} = \beta_{\text{RANSAC}}$	
p-value	0.7732	
Decision	Do not reject H_0	
Conclusion	No significant difference between estimators	

Crisis Period		
	β	$\beta - \beta_{\text{true}}$
OLS	1.3099	-0.1901
RANSAC	1.4792	-0.0208

Hypothesis Test (Crisis)		
Null hypothesis	$H_0 : \beta_{\text{OLS}} = \beta_{\text{RANSAC}}$	
p-value	0.0000	
Decision	Reject H_0	
Conclusion	Estimators differ significantly	

Interpretation of Scenario 3

The comparison between the calm and crisis regimes highlights how market conditions influence the behaviour and reliability of beta estimators.

During the calm period, volatility is low and outliers are extremely rare. Under these ideal conditions, both OLS and RANSAC recover the true beta almost perfectly. Their estimates are statistically indistinguishable, and the regression lines in the left-hand panel nearly overlap. This confirms that when financial returns follow a stable and well-behaved distribution, the classical OLS estimator performs adequately, and robust methods do not provide any substantial improvement.

The situation changes dramatically in the crisis period. Here the return distribution becomes highly volatile, fat-tailed, and heavily contaminated by extreme shocks. OLS reacts strongly to these atypical observations, producing a beta that is significantly biased downward. RANSAC, in contrast, filters out the distorted points and reconstructs a slope much closer to the true value. The hypothesis test confirms that the difference between the two estimators is statistically significant.

Overall, this scenario shows that beta estimation is highly regime-dependent. OLS is reliable only under stable conditions, whereas RANSAC remains accurate even when the data are dominated by extreme and irregular behaviour. This robustness makes RANSAC particularly suitable for financial markets, where crisis periods and outlier clusters are common.

9 Real Data Example: S&P 500 vs. GOOG and TSLA

In this final section, we apply the full methodology to real financial data. We download daily prices for the S&P 500 index and two major stocks—Google (GOOG) and Tesla (TSLA)—over the period 2015–2025. From these price series we compute log-returns and estimate each stock’s beta relative to the market using both OLS and RANSAC.

The choice of GOOG and TSLA is intentional. It follows several preliminary attempts with different stocks aimed at identifying a pair that would clearly illustrate the contrast between the two estimators. GOOG represents a relatively stable, large-cap technology stock with moderate volatility and smooth return dynamics. TSLA, in contrast, displays frequent spikes, abrupt corrections, and a high concentration of extreme daily returns. This combination makes the two assets particularly well suited to the main objective of the empirical exercise: showing how robust methods such as RANSAC respond to real-world financial behaviour compared with traditional OLS.

9.1 GOOG vs. S&P 500

After computing betas and residual standard errors for each stock, we visualise the fitted relationships through scatter plots showing both OLS and RANSAC regression lines. These empirical figures provide an immediate and intuitive comparison of how the two estimators interpret the stock–market relation when confronted with real financial noise, volatility clustering, and irregular return patterns.

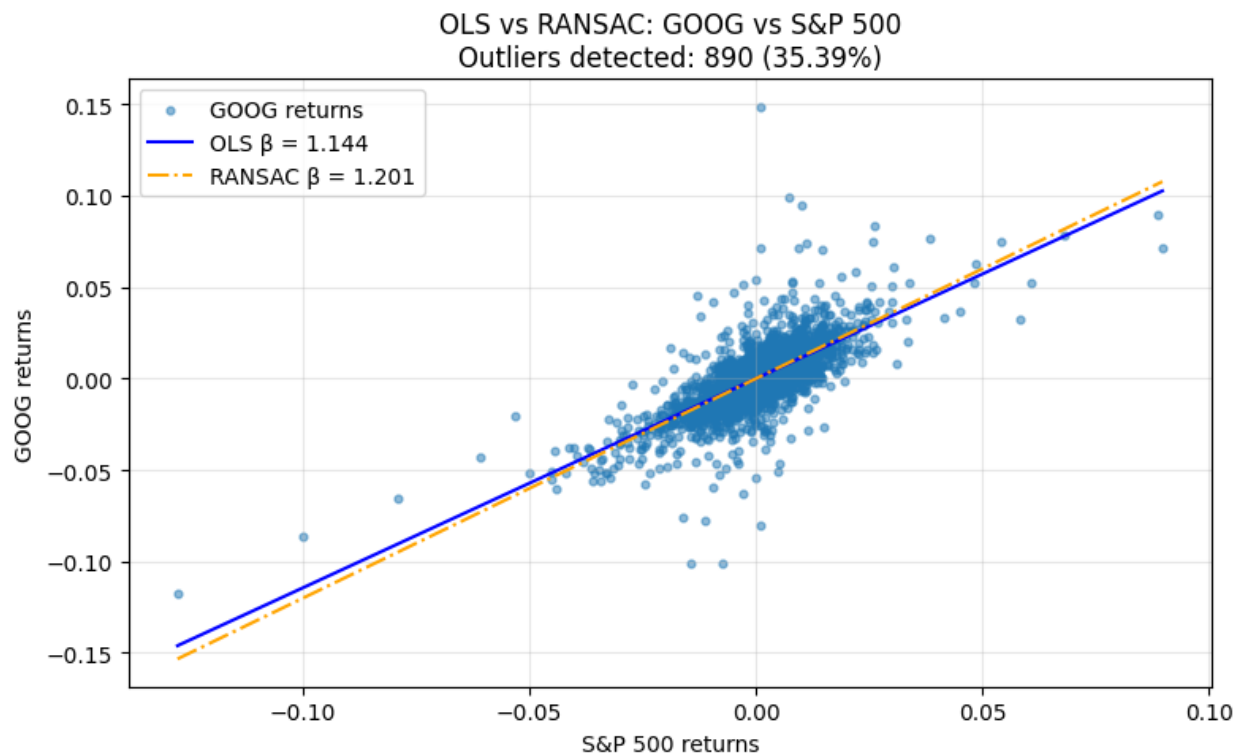


Figure 6: OLS vs. RANSAC beta estimation for GOOG relative to the S&P 500 index. RANSAC detects a substantial number of outliers (35.39%) and produces a beta estimate slightly higher than the OLS estimate.

Table 4: Static Beta Estimates — GOOG vs. S&P 500

Estimator	β	$\beta_{\text{OLS}} - \beta_{\text{RANSAC}}$
OLS	1.1442	-0.0567
RANSAC	1.2008	0.0567

Hypothesis Test

Null hypothesis	$H_0 : \beta_{\text{OLS}} = \beta_{\text{RANSAC}}$
Decision	Do not reject H_0
Conclusion	No significant difference between the estimators

Interpretation of the Real-Data Results: GOOG vs. S&P 500

The comparison between GOOG and the S&P 500 reveals a moderate degree of dispersion in daily returns, with RANSAC identifying 890 observations (approximately 35%) as outliers. This indicates that even relatively stable large-cap technology stocks may exhibit a substantial proportion of atypical movements over long time horizons.

The two beta estimates differ only slightly:

$$\beta_{\text{OLS}} \approx 1.144, \quad \beta_{\text{RANSAC}} \approx 1.201.$$

The lower OLS estimate reflects the influence of extreme observations, which tend to pull the regression line downward. By contrast, RANSAC excludes these contaminating points and focuses on the central structure of the data, producing an estimate more representative of GOOG’s typical co-movement with the S&P 500.

Overall, GOOG exhibits a relatively stable and well-behaved relationship with the market. Although the dataset contains a meaningful number of outliers, their effect on the OLS beta is modest, leading to only a mild downward bias relative to the robust RANSAC estimate.

9.2 TSLA vs. S&P 500

Interpretation of the Real-Data Results: TSLA vs. S&P 500

TSLA shows a completely different pattern from GOOG. RANSAC classifies 1152 observations (approximately 46%) as outliers, meaning that almost half of TSLA’s trading days lie outside the stable linear relationship with the market. This reflects the stock’s well-known high volatility and tendency to experience abrupt price movements.

The discrepancy between the estimated betas is substantial:

$$\beta_{\text{OLS}} \approx 1.503, \quad \beta_{\text{RANSAC}} \approx 1.218.$$

OLS overestimates TSLA’s beta because it is forced to fit episodes of very large positive and negative returns. These extreme values mechanically increase the slope of the regression

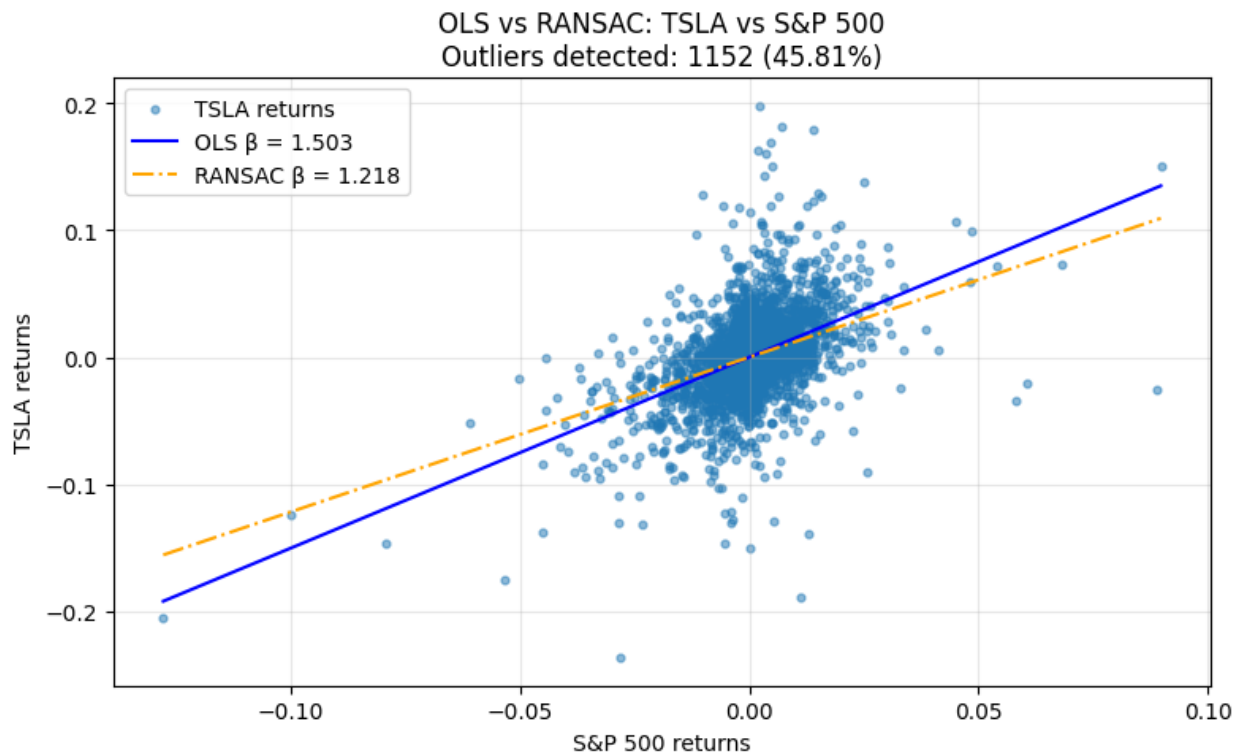


Figure 7: OLS vs. RANSAC beta estimation for TSLA relative to the S&P 500 index. RANSAC identifies a very large fraction of observations (45.81%) as outliers, reflecting TSLA’s highly volatile return dynamics.

Table 5: Static Beta Estimates — TSLA vs. S&P 500

Estimator	β	$\beta_{\text{OLS}} - \beta_{\text{RANSAC}}$
OLS	1.5033	0.2853
RANSAC	1.2181	-0.2853

Hypothesis Test

Null hypothesis	$H_0 : \beta_{\text{OLS}} = \beta_{\text{RANSAC}}$
Decision	Reject H_0
Conclusion	Estimators differ significantly

line, giving the impression of a much higher systematic risk than is typical for the stock. In contrast, RANSAC captures the underlying pattern associated with ordinary trading conditions and returns a significantly lower beta estimate.

Summary The examples of GOOG and TSLA clearly show that the impact of outliers on beta estimation is highly stock-dependent. GOOG, with more stable returns, displays only minor distortions, whereas TSLA—characterised by frequent extreme observations—induces substantial bias in the OLS estimate.

These results reinforce the central message of this project: when return distributions

contain a large number of extreme movements, OLS may fail to provide a reliable measure of systematic risk, while robust estimators such as RANSAC yield more stable and economically meaningful beta values.

9.3 Crisis vs. Calm Period (GOOG vs. S&P 500)

This section compares GOOG's beta in two distinct market regimes:

- **Crisis period (2020–2022):** a highly volatile environment with many outliers, where a robust estimator such as RANSAC is expected to provide a more reliable beta than OLS.
- **Calm period (2017–2019):** a phase with more stable returns, in which OLS and RANSAC betas are expected to be much closer.

For each regime, the code loads GOOG and S&P 500 returns, estimates OLS and RANSAC betas, classifies inliers and outliers, and produces scatter plots that highlight how the two estimators behave under calm versus stressed market conditions.

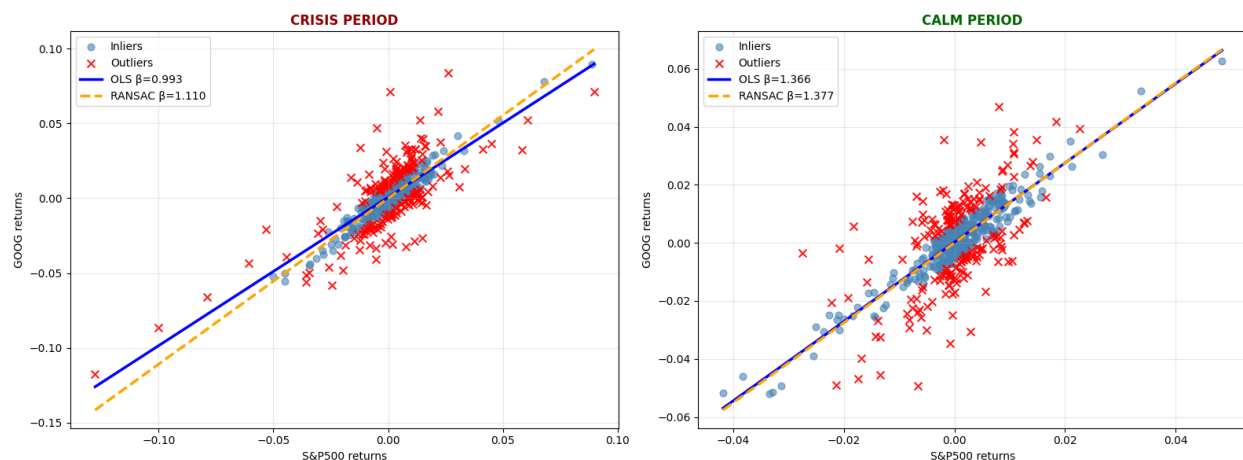


Figure 8: OLS vs. RANSAC beta comparison for GOOG in crisis (left) and calm (right) periods. During the crisis period, OLS is more strongly affected by extreme observations, while RANSAC remains closer to the central relationship. In the calm period, the two estimators deliver very similar slopes.

Crisis vs. Calm Period: Interpretation of Results

The comparison between the crisis period and the calm period highlights how market conditions shape beta estimation, even for a relatively stable large-cap stock such as GOOG.

Table 6: OLS vs. RANSAC β Comparison (GOOG vs. S&P 500) in Crisis and Calm Periods

Crisis period (2020-01-01 – 2022-01-31)		β
OLS		0.9931
RANSAC		1.1095

Hypothesis Test (Crisis)	
Null hypothesis	$H_0 : \beta_{\text{OLS}} = \beta_{\text{RANSAC}}$
p-value	0.0021
Decision	Reject H_0
Conclusion	β_{OLS} and β_{RANSAC} differ significantly

Calm period (2017-01-01 – 2019-01-31)		β
OLS		1.3665
RANSAC		1.3774

Hypothesis Test (Calm)	
Null hypothesis	$H_0 : \beta_{\text{OLS}} = \beta_{\text{RANSAC}}$
p-value	0.8367
Decision	Do not reject H_0
Conclusion	No significant difference between the estimators

Crisis Period (2020–2022) During the crisis, GOOG exhibits a wider and more dispersed return pattern, with several observations deviating from the central cluster. This volatility generates a noticeable gap between the two estimators:

$$\beta_{\text{OLS}} = 0.993, \quad \beta_{\text{RANSAC}} = 1.110,$$

a difference of approximately 12%. OLS is partially pulled downward by negative outliers, whereas RANSAC concentrates on the dense center of the distribution and delivers a higher, more robust beta. This suggests that OLS tends to understate GOOG’s systematic risk when crisis-related noise distorts the data, while RANSAC preserves the underlying slope more effectively.

Calm Period (2017–2019) In the calm window, the cloud of points is tighter and more uniform. As expected, the two estimators align closely:

$$\beta_{\text{OLS}} = 1.366, \quad \beta_{\text{RANSAC}} = 1.377,$$

with a difference of about 1%. Both regression lines almost overlap, indicating that with fewer anomalies OLS behaves reliably, and robust estimation offers only marginal adjustments.

Overall Insight Overall, the results show that:

- In stressed markets, OLS becomes more sensitive to outliers, while RANSAC yields a more stable beta.

- In calm markets, both methods converge because the data contain fewer distortions.

For GOOG, the crisis-induced divergence is smaller than what we observed for TSLA, reflecting GOOG’s lower volatility and more stable risk profile.

10 Conclusion

The analysis conducted throughout this project highlights a central point in empirical finance: beta estimation is highly sensitive to outliers, and this sensitivity varies considerably across market regimes. The combination of simulation experiments, real stock data (GOOG and TSLA), and robust estimation techniques provides a coherent and consistent picture of how traditional and robust methods behave under different market conditions.

The simulation results show that outliers are not merely accidental anomalies but structural features of financial returns. When 10% of observations are contaminated, OLS deviates sharply from the true beta, whereas RANSAC remains stable and much closer to the underlying relationship. When the fraction of outliers falls to 0.25%, the two estimators converge, confirming that the issue lies in data contamination rather than in the linear model itself. These controlled experiments underscore how sensitive OLS can be to extreme observations and how effectively RANSAC isolates the core structure of the data.

The comparison between calm and crisis market periods further reinforces this insight. In crisis environments, returns are affected by shocks, jumps, and irregular fluctuations, which mechanically distort OLS estimates. RANSAC, by filtering out anomalous points, provides a more reliable and stable measure of systematic risk. Conversely, in calm markets—such as GOOG’s 2017–2019 window—extreme observations are fewer, and the difference between the two estimators becomes negligible. This confirms that robust methods become essential primarily under volatile or turbulent market conditions.

The empirical evidence from real stocks supports the conclusions drawn from the simulations. GOOG, a relatively stable large-cap stock, shows only moderate differences between OLS and RANSAC estimates. TSLA, which is far more volatile and characterized by frequent extreme returns, exhibits large divergences between the two estimators, particularly during stressed periods. This contrast illustrates how the intrinsic volatility of an asset determines its sensitivity to outlier-driven distortions in beta estimation.

Relying exclusively on OLS may therefore lead to misestimation of systematic risk, with potential consequences for portfolio risk assessment, asset pricing, hedging strategies, and performance evaluation. Robust methods such as RANSAC offer a more dependable measure of beta in the presence of noise, jumps, or structural anomalies, without sacrificing efficiency when the data are well behaved.

10.1 Final Consideration

Beta is not only a property of the stock itself but also a reflection of the data environment in which it is estimated. Traditional OLS remains appropriate in stable conditions, but robust estimators become essential when returns contain shocks or irregular behaviour. Combining classical and robust techniques provides a more complete and accurate understanding of systematic risk, both in theory and in practical financial applications.

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